

FIFA 21 PLAYER MARKET & PERFORMANCE ANALYSIS –

Bussiness Problem Statement

“Professional football clubs invest millions of euros every season in player transfers and salaries. Identifying players who can deliver high performance while providing long-term value is one of the biggest challenges in modern football management. Recruitment decisions are influenced by several factors, including player age, overall performance, market value, wages, playing position, and future potential. Without data-driven analysis, clubs may overpay for players, overlook promising young talent, or make financially inefficient transfer decisions.

This project analyzes the FIFA 21 Player Dataset to identify the key factors that influence player market value, wages, and future potential. Through data cleaning, exploratory data analysis (EDA), and visualization, the project aims to uncover meaningful patterns in player performance, salary distribution, club characteristics, and talent development. The insights generated will support data-driven recruitment strategies, financial planning, and player investment decisions, enabling football clubs to maximize their return on investment while building stronger and more competitive teams.”

Deliverables (Project Workflow)

1. Data Import & Initial Exploration

- Import the FIFA 21 Player Dataset into Python.
- Explore the dataset structure, dimensions, data types, and completeness.
- Perform an initial review of important variables such as player name, age, nationality, club, position, overall rating, potential, market value, wages, and release clause.
- Identify missing values, duplicate records, and data inconsistencies.

2. Data Cleaning & Preparation

- Handle missing values and duplicate records.
- Clean and standardize currency columns such as Market Value, Wage, and Release Clause.
- Convert monetary values from text format (€, K, M) into numerical values.

- Correct inconsistent data types and formatting issues.
- Create derived features such as:
 - Potential Growth (Potential – Overall Rating)
 - Value per Rating
 - Wage-to-Value Ratio
- Remove irrelevant or redundant columns to prepare the dataset for analysis.

3. Data Transformation & Organization

- Organize the cleaned dataset for analytical tasks.
- Convert categorical variables into appropriate formats where required.
- Create additional analytical features to improve business insights.
- Ensure the dataset is structured and optimized for exploratory analysis and visualization.

4. Exploratory Data Analysis (EDA)

- Identify the factors that influence player market value.
- Analyze the relationship between age, overall rating, potential, and market value.
- Compare player performance across different positions, clubs, and nationalities.
- Identify players with the highest growth potential.
- Analyze salary distribution across different playing positions.
- Discover trends that can support player recruitment and investment decisions.

5. Visualization & Dashboard Development

- Develop visualizations to communicate key insights effectively.
- Create charts showing player age distribution, market value distribution, salary trends, and position-wise analysis.
- Compare club performance and nationality representation.
- Visualize relationships between player ratings, wages, and market value.
- Highlight business insights that support strategic recruitment and financial planning.

6. Reporting & Documentation

- Prepare a detailed Data Cleaning Report.
- Document all exploratory analysis and visualization findings.
- Summarize key trends, observations, and business recommendations.
- Present conclusions using charts, tables, and descriptive narratives suitable for stakeholders.

7. Project Packaging & Presentation

- Organize the dataset, Jupyter Notebook, visualizations, reports, and presentation into a structured project folder.
- Prepare a professional PowerPoint presentation highlighting the business problem, methodology, findings, and recommendations.
- Develop a portfolio-ready project suitable for internship evaluation, GitHub, LinkedIn, and placement interviews.

Business Questions

The following business questions will guide the analysis:

Player Market Analysis

- Which factors have the greatest impact on a player's market value?
- Does a higher overall rating always result in a higher market value?
- How does player age influence market value?

Talent Identification

- Which young players have the highest future potential?
- Which players appear undervalued based on their current ratings?
- Which positions provide the best value for investment?

Salary Analysis

- Which playing positions receive the highest average wages?
- Is there a relationship between player wages and overall performance?

Clubs & Nationality Analysis

- Which clubs have the highest average player ratings?
- Which countries produce the highest-rated players?
- What are the overall trends in player distribution across positions and nationalities?